

Pool Boiling Enhancement with High Entropy Oxides (HEOs)

Abstract:

This study explores the potential of high entropy oxides (HEOs) as pool boiling additives in deionized (DI) water to enhance boiling heat transfer. Experiments were conducted using spinel $[(\text{Al}_{1/6}\text{Co}_{1/6}\text{Cr}_{1/6}\text{Fe}_{1/6}\text{Mn}_{1/6}\text{Ni}_{1/6})_3\text{O}_4]$, perovskite $[\text{La}(\text{Co}_{0.2}\text{Cr}_{0.2}\text{Fe}_{0.2}\text{Mn}_{0.2}\text{Ni}_{0.2})\text{O}_3]$, $\text{Y}(\text{Co}_{0.2}\text{Cr}_{0.2}\text{Fe}_{0.2}\text{Mn}_{0.2}\text{Ni}_{0.2})\text{O}_3$, and rock salt $[(\text{Mg}_{0.2}\text{Co}_{0.2}\text{Ni}_{0.2}\text{Li}_{0.2}\text{Zn}_{0.2})\text{O}]$ structured HEOs. A significant enhancement in critical heat flux (CHF) and heat transfer coefficient (HTC) was observed. Specifically, 0.05 wt% $[\text{La}(\text{Co}_{0.2}\text{Cr}_{0.2}\text{Fe}_{0.2}\text{Mn}_{0.2}\text{Ni}_{0.2})\text{O}_3]$ yielded a 66.1% increase in CHF, while 0.05 wt% $\text{Y}(\text{Co}_{0.2}\text{Cr}_{0.2}\text{Fe}_{0.2}\text{Mn}_{0.2}\text{Ni}_{0.2})\text{O}_3$ improved CHF by 62.5% and HTC by 136.5% compared to the DI water baseline on plain copper surfaces. To investigate dispersion behavior and long-term colloidal stability, a 15-day ultrasonication-based dispersion test was conducted in DI water. The study identified key mechanisms for effective dispersion, including deagglomeration, formation of hydroxyl functional groups, and electrostatic repulsion. These findings support the potential of HEOs as scalable additives for enhancing thermal fluids in pool boiling applications.

Experimental setup:

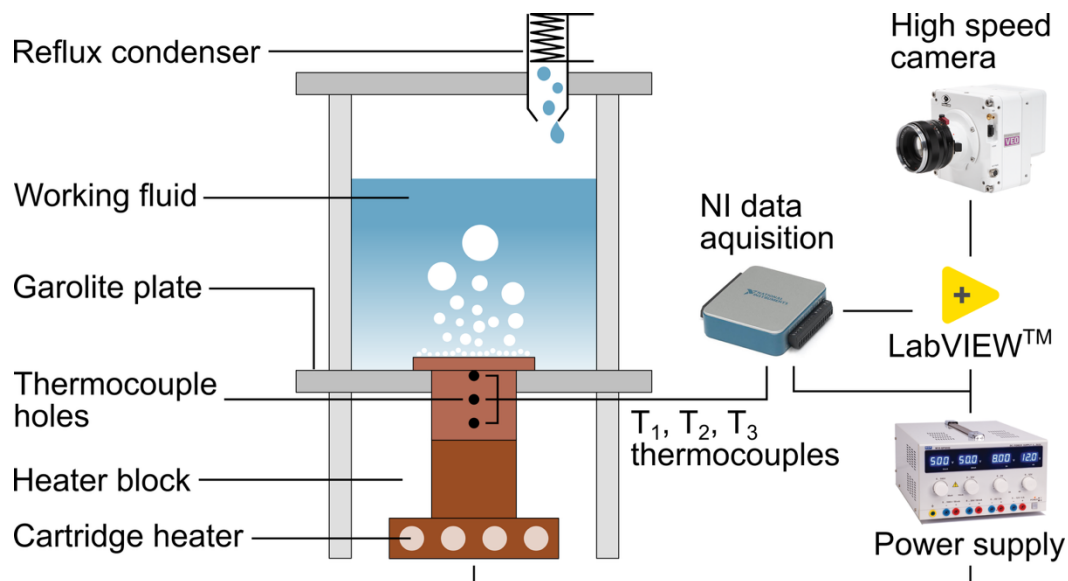


Fig 1: Schematic of the pool boiling setup along with thermocouple location for temperature measurements.

Results:

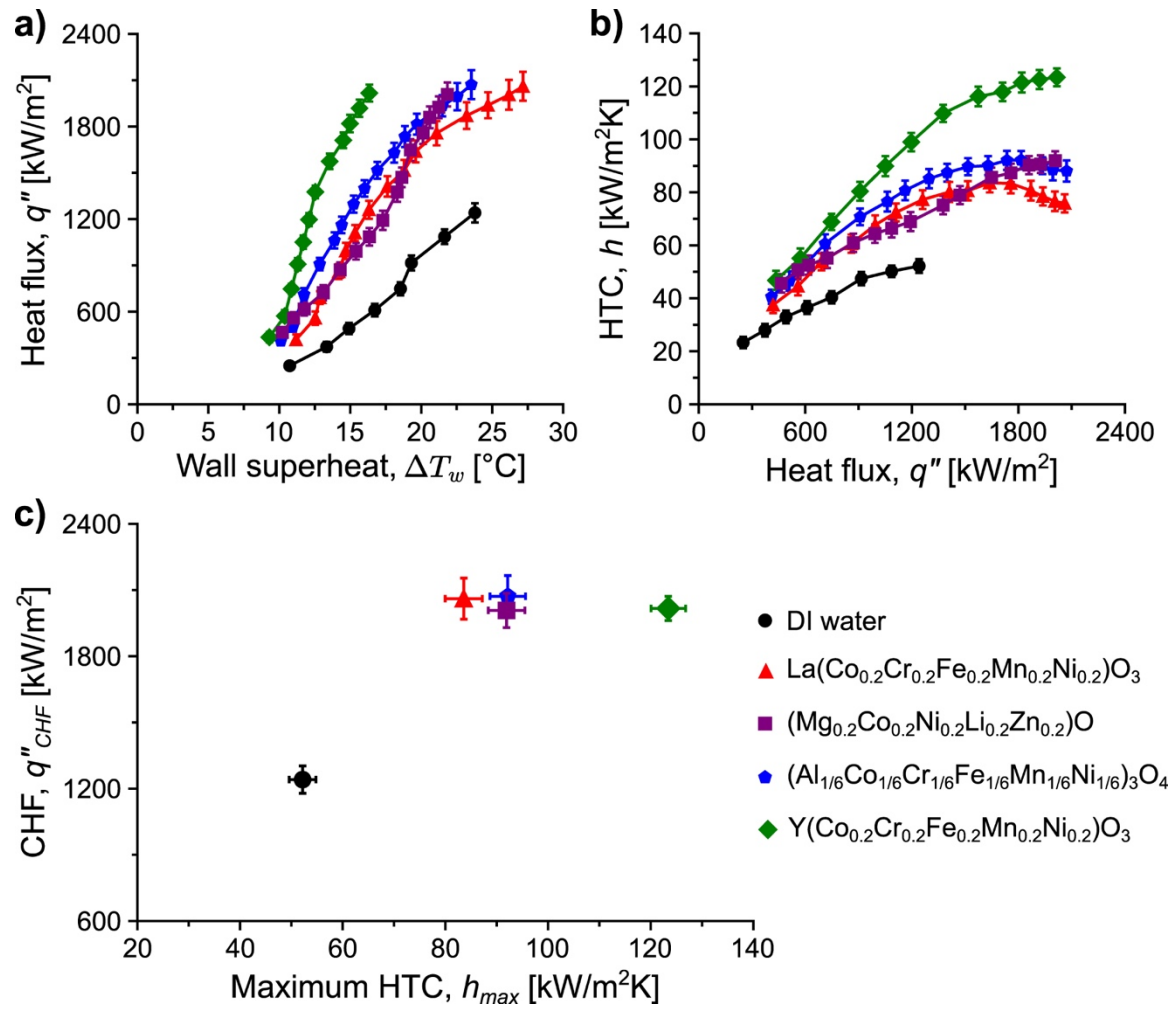


Fig 2: Boiling performance of HEO dispersions compared to DI water (a) heat flux vs wall super heat, (b) HTC vs heat flux, and (c) CHF vs maximum HTC (h_{max}).